Relational Databases

COM 736 Database Systems and Data Analytics

Objectives

Database Models	• Brief Overview
Relational Model	• History and data storage

Database

- A Database is an **integrated collection** of stored operational data used by the application systems of a particular enterprise.
  - Data for all applications in the enterprise is stored in the integrated database **not** individual files.
  - **Centralised control** of its operational data
  - **Data independence**

Database Models

Hierarchical Model and Network Model	• Early models, closely associated with physical storage and programs are written to interact with the data
Relational Model	• Conceptual model and physical storage and separated with a Structured Query Language to interact through DMBS
Object Oriented Model	• Dealing with complex objects using object oriented model
XML Document Model	• eXtended Mark-up Language for web applications based on document models and relational models
Big Data Systems and NoSQL Database Model	• Fulfilling the requirements of non traditional data type storage and retrieval, fast access of enormous data through various applications and platforms.

The Relational Model

- In Relational model relations hold the data
  - A Relation is a two dimensional Table
  - Data is organised into tables.
    - A row corresponds to a record
    - A column corresponds to a field, also known as attribute

The Relational Model

Tables are different from sequential files

- Each table contains only one type of record
- Each record in a table has the same number of fields
- The order of the records within a table has no significance

The Relational Model

Customer : Table

Attributes

Records

CustomerNumber	Name	Address	Type
5567	Jones	Cross Rd	Business
2913	Brown	Cross Rd	Personal
4890	Jones	Main St	Business

The Relational Model

- Each row of a table is **uniquely identified** by the data values from one or more columns.
  - Known as **Primary Key**
    - In this customer table, Number is the primary key

CustomerNumber	Name	Address	Type
5567	Jones	Cross Rd	Business
2913	Brown	Cross Rd	Personal
4890	Jones	Main St	Business

  - If there are more than one possible way of uniquely identifying each record of a table, each known as a Candidate key.
    - One of the candidate keys must be selected as the primary key

The Relational Model

- The relationships between rows of data in different tables are represented solely by the data values of one or more column.

COURSE

Course_ID	Course_Name	Level
BSC_COMP	BSc Computing	Undergraduate
MSC_COM_SCI	MSc Computer Science	Postgraduate
MSC_COM_NW	MSc Computer Science with Networking	Postgraduate
MSC_COM_BDA	MSc Computer Science with Big Data Analytics	Postgraduate

STUDENT

Student_ID	First_Name	Last_Name	Course_ID
S001	Jack	Adams	BSC_COMP
S002	Jill	Jones	MSC_COM_SCI
S003	Bob	Who	MSC_COM_SCI
S004	Alice	Davies	MSC_COM_BDA

The Relational Model

- A field referring to the primary key field of another table is a foreign key.

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Primary key – Course_ID

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S004	Alice	Davies	MSC_COM_BDA

Primary Key – Student_ID

Foreign key – COURSE_ID

Summary

- Database
- Database Models
- Relational Model
- Next:
  - Database Design: Entity Relationship modelling